

HR ANALYTICS

DATA DICTIONARY

CodeAlpha Internship Project • Manya Sain

This document defines the fields, supporting data files, business meaning, and analytical use of the data used in the HR Analytics dashboard.

Repository: [ManyaSain/CodeAlpha_HR_Analytics_Dashboard](#)

1. Data Dictionary — Employee.csv

Employee.csv is the primary employee/workforce table. The fields below describe the employee information used for HR analysis and Power BI reporting.

Column / Field	Data Type	Description	Dashboard / Analytical Use
Employee ID	Integer / ID	Unique identifier assigned to each employee.	Employee identification, counting and data relationships.
Age	Integer	Age of the employee in years.	Average age and workforce segmentation.
Gender	Categorical	Gender category recorded for the employee.	Gender-wise workforce and attrition analysis.
Department	Categorical	Department in which the employee works.	Department filters and attrition comparison.
Job Role	Categorical	Specific role or position held by the employee.	Job-role analysis and attrition comparison.
Education	Categorical / ID	Employee education information or related reference value.	Education-based workforce analysis.
Education Level	Categorical / ID	Standardized education-level category or reference.	Education segmentation and dimension relationship.
Job Level	Integer / Categorical	Employee's organizational or job level.	Workforce segmentation and HR analysis.
Monthly Income / Salary	Numeric	Employee compensation value recorded in the dataset.	Salary analysis and workforce comparison.
Years at Company	Integer	Number of years the employee has been with the organization.	Tenure analysis.
Years in Current Role	Integer	Number of years spent in the current role.	Role-tenure analysis.
Years Since Last Promotion	Integer	Number of years since the employee's last promotion.	Career progression analysis.
Years With Current Manager	Integer	Number of years working with the current manager.	Manager and tenure analysis.
Overtime	Categorical	Indicates whether the employee works overtime.	Workload-related workforce analysis.
Job Satisfaction	Rating / ID	Employee job-satisfaction rating or linked satisfaction level.	Satisfaction distribution and satisfaction vs attrition.
Performance Rating	Rating / ID	Performance or management rating associated with the employee.	Performance vs attrition analysis.
Manager Rating	Rating / ID	Rating associated with manager assessment.	Manager rating vs attrition analysis.
Attrition	Categorical (Yes/No)	Indicates whether the employee has left the organization.	Primary field for attrition analysis.
Hire Date	Date	Date on which the employee joined the organization.	Hiring trend and time-based analysis.

2. Supporting Dimension / Reference Files

These files support the employee table and help standardize ratings, satisfaction, education levels and time-based analysis.

File	Purpose	Typical Content / Key
EducationLevel.csv	Education-level lookup/reference table used to standardize education categories.	Education Level ID and level/name.
PerformanceRating.csv	Reference data for performance or management rating levels.	Rating ID and rating description.
RatingLevel.csv	Lookup table defining rating levels used by the model.	Rating ID, level and description.
SatisfiedLevel.csv	Reference table for employee satisfaction levels.	Satisfaction ID, level and description.
DimDate.txt	Date dimension used for time-based analysis and trend calculations.	Date, year, month, quarter and related date attributes.

3. Data Model Concept

The project combines the primary employee table with supporting lookup/reference files. Corresponding IDs or keys can be used to connect employee records with education, performance, rating and satisfaction information. DimDate provides a dedicated date structure for time-based reporting and hiring trends.

4. Business Fields

Business Concept	Main Field(s)	Purpose
Workforce Size	Employee ID	Determines the number of employees in the workforce.
Attrition	Attrition	Core HR outcome analyzed throughout the dashboard.
Organization	Department, Job Role, Job Level	Shows where employees work and how attrition is distributed.
Employee Experience	Job Satisfaction	Helps examine employee satisfaction patterns.
Performance	Performance Rating, Manager Rating	Supports performance and management comparisons with attrition.
Diversity	Gender	Supports gender-wise workforce and attrition views.
Tenure	Years at Company, Years in Current Role	Helps understand employee and role tenure.
Career Growth	Years Since Last Promotion	Supports career-progression analysis.
Hiring Trend	Hire Date, DimDate	Supports historical trend and forecast analysis.

5. Dashboard Measures / Metrics

Metric	Definition / Purpose
Total Employees	Count of employee records representing the workforce.
Active Employees	Count of employees classified as active, where an active-status field is available.
Attrition Count	Count of employees whose Attrition status is Yes.
Attrition Rate	Attrition count divided by the relevant workforce count, expressed as a percentage.
Average Age	Average age of employees in the dataset.
Average Salary	Average salary or income value in the employee dataset.
Hiring Trend	Number of employees hired across time periods using the date dimension.
Forecast	Power BI estimate of future values based on historical time-series behavior.

6. Data Quality & Usage Notes

- Employee ID should remain unique for employee-level analysis.
- Categorical values such as Department, Job Role, Gender and Attrition should be standardized before analysis.
- Rating and satisfaction IDs should be interpreted together with their corresponding lookup tables when applicable.
- Date fields should use a proper Date data type and should be connected to DimDate for time intelligence.
- Numeric fields such as Age, Salary and tenure should remain numeric for correct aggregation.
- Forecast outputs are estimates based on historical patterns and should be treated as planning guidance.
- Relationships between satisfaction, performance and attrition indicate patterns or associations; they do not by themselves prove causation.

7. Repository Data Assets

The data files documented here correspond to the project repository assets: Employee.csv, EducationLevel.csv, PerformanceRating.csv, RatingLevel.csv, SatisfiedLevel.csv and DimDate.txt. The repository also contains the Power BI report, dashboard preview, README.md and .gitignore.

8. Document Information

Project: CodeAlpha HR Analytics Dashboard
Prepared By: Manya Sain
Program: B.Tech Information Technology
Institution: Guru Jambheshwar University of Science & Technology (GJUST), Hisar
Primary Tool: Microsoft Power BI
Purpose: Internship / Portfolio Project